

Treasury Inflation-Protected Securities (TIPS)

The United States began issuing Treasury Inflation-Protected Securities (TIPS) in 1997. Backed by the full faith and credit of the US government and assurances that inflation cannot eat away at their value, TIPS provide a risk-free asset for US-based investors.

The face value and coupon payments are both indexed to keep pace with inflation and preserve purchasing power, and their yields are quoted in real inflation-adjusted terms. Whenever positive inflation (as opposed to deflation) is expected, real yields will be less than the nominal yields quoted on traditional (i.e., not inflation adjusted) bonds. As an approximation:

$$\text{real interest rate} = \text{nominal interest rate} - \text{expected inflation rate}$$

Or, more precisely, the exact formula to relate these variables is:

$$(1 + \text{nominal interest rate}) = (1 + \text{real interest rate}) \times (1 + \text{expected inflation rate})$$

Nominal interest rates are determined by compensation expected to keep pace with inflation plus a real rate of return for the investor. Of course, supply and demand affect bond prices and interest rates. Real interest rates can be negative.

Investors may expect a positive nominal return on their investment (otherwise, there is no reason to invest), but that return may not be able to keep pace with inflation. Unlike traditional bonds, TIPS yields are quoted as real interest rates.

Their nominal yields are not known in advance, because they depend on the subsequently realized inflation experience. Conversely, we know nominal yields for traditional bonds, but their real yields can only be known after observing the realized path of inflation up to the maturity date.

Inflation adjustments for TIPS are linked to the Consumer Price Index for All Urban Consumers (CPI-U). These adjustments are tracked in terms of the “accrued principal,” which is a unique term for TIPS. Accrued principal